AI-Orchestrated Employee Onboarding Training Module

Bayesian Hyperparameter Optimisation of a Fine-Tuned Small Language Model with Retrieval-Augmented Generation

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Abstract

Employee onboarding is a critical process in organisations, yet traditional training methods are often time-consuming, inconsistent, and difficult to scale. This thesis investigates a practical problem: general-purpose language models perform poorly in domain-specific enterprise training contexts, and fine-tuning them optimally under hardware constraints is a resource-intensive, poorly understood challenge.

The central research question is: *Can Bayesian hyperparameter optimisation of a fine-tuned small language model improve the performance of an AI-driven onboarding training module compared to non-optimised baseline configurations?*

To address this question, the study applies a design research methodology, developing and evaluating an artefact: an AI-orchestrated onboarding training module. The module integrates the Qwen 2.5 1.5B SLM fine-tuned using QLoRA (Quantised Low-Rank Adaptation) on a mixed corpus of four datasets totalling approximately 4,000 examples - OpenHermes-2.5, UltraChat 200k, ConvoLearn, and a custom synthetic onboarding Q&A dataset. Bayesian optimisation via the Optuna framework is applied to search the space of learning rate, LoRA rank, and LoRA dropout over ten trials. The optimised model is embedded in a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB for domain-grounded inference.

The Bayesian optimisation identified optimal hyperparameters achieved an evaluation loss of 0.993, a 4.1% improvement over the next-best non-optimised configuration. Final fine-tuning with these parameters achieved a training loss of 0.967 and mean token accuracy of 82.2%, demonstrating that higher learning rates and higher LoRA ranks outperform conservative configurations under data-scarce conditions. Qualitative evaluation confirmed appropriate in-domain responses, correct out-of-domain refusal, and coherent multi-turn behaviour.

These findings demonstrate that systematic Bayesian hyperparameter optimisation provides measurable, reproducible improvement over arbitrary configurations in fine-tuning small language models for domain-specific applications. The proposed SLM-plus-RAG architecture provides a scalable, computationally feasible framework for AI-driven enterprise onboarding that is accessible without high-end infrastructure.

Acknowledgements

I would like to express my sincere gratitude to my thesis supervisor for their guidance, patience, and constructive feedback throughout this project. Their insights into machine learning and natural language processing were invaluable in shaping the direction of this research.

I am also grateful to the open-source community behind Hugging Face, Optuna, LangChain, and ChromaDB, whose publicly available tools and datasets made this work feasible. Without the accessibility of pre-trained models, optimisation frameworks, and high-quality conversational datasets, an undertaking of this scope would not have been possible on consumer-grade hardware.

Finally, I thank my family and friends for their unwavering support and encouragement throughout my studies.

List of Abbreviations

Abbreviation	Definition
AI	Artificial Intelligence
API	Application Programming Interface
BLEU	Bilingual Evaluation Understudy
ChromaDB	Chroma Vector Database
CPU	Central Processing Unit
EI	Expected Improvement
FP16	16-bit Floating Point
GPU	Graphics Processing Unit
LLM	Large Language Model
LoRA	Low-Rank Adaptation
LSTM	Long Short-Term Memory
NF4	4-bit NormalFloat Quantisation
NLP	Natural Language Processing
PDF	Portable Document Format
PEFT	Parameter-Efficient Fine-Tuning
Q&A	Question and Answer
QLoRA	Quantised Low-Rank Adaptation
RAG	Retrieval-Augmented Generation
RNN	Recurrent Neural Network
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
SFT	Supervised Fine-Tuning
SLM	Small Language Model
TPE	Tree-structured Parzen Estimator
TRL	Transformer Reinforcement Learning (library)

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1

Introduction

Employee onboarding is a critical process in modern organisations. Research consistently shows that effective onboarding improves employee retention, accelerates time-to-productivity, and strengthens organisational culture [3]. Yet traditional training methods, static documentation, instructor-led sessions, and manually curated materials, frequently suffer from three structural weaknesses. They lack personalisation to the individual learner's pace and prior knowledge, they are difficult to scale as organisations grow, and their quality is inconsistent across delivery instances.

Recent advances in large language models (LLMs) and small language models (SLMs) offer new opportunities to address these challenges. Interactive AI systems can generate explanations, answer questions, and provide contextual guidance in real time, potentially transforming onboarding from a one-time event into an adaptive, ongoing process [4]. However, deploying such systems in enterprise settings raises immediate practical problems: general-purpose models are not trained on organisational knowledge, fine-tuning them requires careful hyperparameter selection, and running or training large models demands infrastructure that many organisations do not possess.

This thesis addresses these challenges by developing and evaluating an AI-orchestrated onboarding training module built around a fine-tuned small language model augmented with Retrieval-Augmented Generation (RAG). The system integrates the Qwen 2.5 1.5B SLM, trained on a mixed conversational corpus, with a ChromaDB vector store that provides domain-specific context at inference time. To maximise training efficiency under hardware constraints, Bayesian hyperparameter optimisation via Optuna is applied to the fine-tuning process.

1.1 Problem

The problem underlying this thesis is as follows. Organisations increasingly explore AI-driven training systems to accelerate employee onboarding and knowledge transfer. However, the deployment of such systems faces two interconnected challenges that have not been well addressed in practice.

First, general-purpose language models, whether large or small, are not pre-trained on domain-specific organisational knowledge. They may generate plausible but factually incorrect or contextually inappropriate responses when queried about internal policies, procedures, or training content. Fine-tuning on domain-relevant data addresses this problem but introduces a second challenge.

Second, selecting optimal hyperparameters for fine-tuning is a resource-intensive, non-trivial task. The quality of the fine-tuned model is highly sensitive to choices such as learning rate, adapter rank, and regularisation strength [13]. Traditional approaches like manual tuning, grid search, and random search are computationally expensive and yield inconsistent results. This problem is compounded when hardware constraints (such as the consumer-grade GPUs available in this study) limit the feasible number of training runs. There is therefore a need for efficient, principled optimisation methods that can identify high-performing configurations with a small number of trials.

This problem is of general interest to any organisation seeking to deploy AI-driven training systems without access to enterprise-grade infrastructure. The relevance of the problem is further underscored by recent evidence that SLMs are increasingly capable of handling specialised tasks when properly fine-tuned, while remaining deployable on standard hardware [4, 15].

1.2 Research Question

The research question in this thesis is:

Can Bayesian hyperparameter optimisation of a fine-tuned small language model improve the performance of an AI-driven onboarding training module compared to non-optimised baseline configurations?

This research question is innovative in that it specifically targets the *intersection* of three components, SLM fine-tuning, Bayesian optimisation, and RAG-based enterprise deployment. The lack of onboarding datasets and AI-specific research shows that this has not been systematically evaluated in the onboarding domain. The question is practically motivated (enterprise onboarding is a concrete, widespread problem) and theoretically grounded (Bayesian optimisation is a well-established

technique whose application to SLM fine-tuning under hardware constraints is underexplored). Answering it advances knowledge about the effectiveness of Bayesian optimisation in this context.

1.3 Sub-Question

The sub-question complementing the main research question is:

How does the Bayesian optimised model compare to non-optimised configurations in terms of response quality, domain relevance, out-of-domain refusal, and multi-turn consistency within the onboarding training module?

This sub-question directs attention to the qualitative and practical dimensions of the system's performance. These are dimensions that numerical loss metrics alone cannot capture. It is answered through a structured qualitative evaluation of model outputs across representative query types.

2

Scientific Grounding

This chapter provides the scientific grounding required to situate this thesis within the research literature. Section 2.1 introduces large and small language models and their architectures. Section 2.2 reviews fine-tuning approaches with particular attention to parameter-efficient methods. Section 2.3 introduces Retrieval-Augmented Generation. Section 2.4 develops the theoretical basis for Bayesian learning and Bayesian optimisation. Section 2.5 critically reviews previous solutions and identifies the gap this thesis addresses.

2.1 Large Language Models and Small Language Models

Large Language Models (LLMs) are a class of machine learning models trained on large-scale text corpora to learn statistical patterns, linguistic structures, and semantic relationships in natural language. Most modern LLMs are built upon the Transformer architecture, introduced by Vaswani et al. [24], which enables capturing of long-range dependencies in sequences through self-attention mechanisms. For each token, self-attention computes a weighted combination of all other tokens in the sequence, allowing the model to selectively attend to contextually relevant parts of the input.

LLMs are typically trained via next-token prediction on massive web-scale corpora, a self-supervised objective that requires no labelled data [7]. Prominent examples include GPT-4 by OpenAI and the Qwen family by Alibaba. These models have demonstrated strong capabilities across text generation, question answering, summarisation, and conversational interaction [20].

There is a key distinction relevant to this thesis between LLMs and Small Language Models (SLMs). Belcak et al. [4] define an SLM as a language model capable of running on consumer hardware with sufficiently low latency for practical agentic use. In 2025, this approximately corresponds to models with fewer than 10 billion parameters. The authors argue that SLMs are not only more economical than LLMs but are *sufficiently powerful* for a wide range of specialised, repetitive, and domain-bounded tasks, making them the appropriate choice for enterprise applications where privacy, cost, and latency matter.

The Qwen 2.5 1.5B model used in this thesis is an SLM with 1.5 billion parameters [19]. It is part of the Qwen 2.5 family, which spans from 0.5B to 72B parameters, and has been shown to achieve competitive performance on instruction-following and conversational tasks relative to its size. Its publicly available weights, open licence, and demonstrated fine-tunability on consumer GPUs make it well-suited for this study.

2.2 Fine-Tuning Language Models

Fine-tuning adapts a pre-trained language model to a specific task or domain by continuing training on a curated dataset [10]. While pre-trained models encode broad linguistic knowledge, they lack domain-specific information and may produce contextually inappropriate outputs in specialised settings. Fine-tuning addresses this by shifting model behaviour toward the target domain.

Several fine-tuning paradigms exist. *Full fine-tuning* updates all model parameters and can yield strong performance but requires substantial GPU memory and compute. This is impractical at scale for models with billions of parameters. *Instruction tuning* fine-tunes the model specifically on prompt–response pairs, improving instruction-following capabilities without necessarily specialising the model's factual knowledge [11]. *Parameter-efficient fine-tuning* (PEFT) methods update only a small subset of parameters, dramatically reducing training cost [10]. In this case, PEFT was performed using QLoRA.

2.2.1 Low-Rank Adaptation (LoRA) and QLoRA

Low-Rank Adaptation (LoRA), introduced by Hu et al. [13], is the most widely adopted PEFT method. LoRA freezes all pre-trained model weights and injects trainable low-rank decomposition matrices into the attention layers of the Transformer. For a weight matrix $W_0 \in \mathbb{R}^{d \times k}$, the weight update is decomposed as:

$$W = W_0 + \Delta W = W_0 + BA \tag{2.1}$$

where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ are trainable matrices of rank $r \ll \min(d, k)$. Only A and B are trained; W_0 remains frozen. This reduces the number of trainable parameters from $d \times k$ to $r(d + k)$, which for typical attention matrices with $r = 12$ represents a reduction of over 99%.

Dettmers et al. [9] extended LoRA with 4-bit NormalFloat (NF4) quantisation (QLoRA), loading the frozen base model in 4-bit precision while training the LoRA adapters in 16-bit. This further reduces GPU memory consumption: the Qwen 2.5 1.5B model occupies approximately 3.5 GB in 4-bit, compared to 3 GB in full float16 and 6 GB in float32. QLoRA has been shown to recover near-full fine-tuning performance while being accessible on consumer GPUs, making it the dominant choice for resource-constrained fine-tuning.

A critical design decision in LoRA is the choice of rank r: higher ranks allow more expressive adaptations but increase trainable parameter count and risk overfitting on small datasets. The LoRA dropout rate p applies dropout to the LoRA outputs as regularisation. Both r and p are treated as optimisable hyperparameters in this thesis.

2.2.2 Training Data for Fine-Tuning

The quality and composition of fine-tuning data significantly affect model performance. This thesis uses four datasets, each chosen for a specific purpose.

UltraChat 200k [11] is a filtered subset of the UltraChat dataset, which contains 1.5 million high-quality multi-turn instructional dialogues generated by GPT-based models. Ding et al. demonstrate that models fine-tuned on UltraChat substantially outperform those trained on smaller or less diverse instruction datasets in conversational settings. The subset contains 5% data with grammatical errors to take poor quality prompts into account. It also helps in removing responses containing a lack of emotion or opinions. For this thesis, the dataset was shuffled and 1000 rows were selected.

OpenHermes-2.5 [23] is a large, diverse collection of over one million synthetic instruction-following conversations compiled from multiple sources, covering diverse topics and interaction styles. It is widely used as a general-purpose instruction-tuning dataset in the open-source community. For this thesis, the dataset was shuffled and 2000 rows were selected.

ConvoLearn [21] is a dataset of teacher–student conversational exchanges specifically designed for educational and tutoring contexts, making it particularly relevant for an onboarding tutoring system. For this thesis, the dataset was shuffled and 1000 rows were selected.

Synthetic Onboarding Q&A is a custom dataset of approximately 520 question–answer pairs covering organisational policies and onboarding procedures, generated

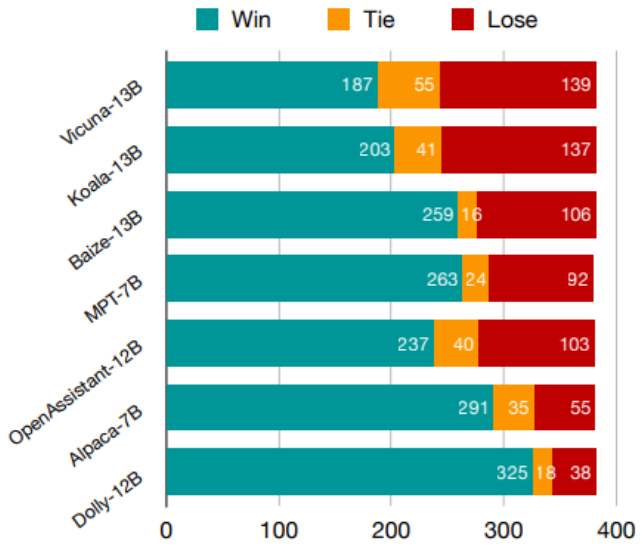


Figure 2.1: Response comparison of an SLM trained with UltraChat 200k with other baseline models [11]

using ChatGPT and formatted to match the UltraChat structure. This dataset provides the domain-specific grounding that the other general-purpose datasets cannot supply.

2.3 Retrieval-Augmented Generation

A fundamental limitation of any fine-tuned language model is that its knowledge is bounded by its training data. For an onboarding system, this means the model cannot answer questions about specific internal documents, policies, or training materials that postdate or fall outside its training corpus. Retrieval-Augmented Generation (RAG), introduced by Lewis et al. [16], addresses this by augmenting the model with a retrieval component that fetches relevant document chunks at inference time.

In a RAG pipeline, a query is embedded into a dense vector representation and used to retrieve the most semantically similar chunks from a pre-indexed document store. The retrieved chunks are concatenated with the query to form an augmented prompt, which is passed to the language model. This allows the model to ground its responses in up-to-date, domain-specific documents without requiring those documents to be memorised during training.

Gao et al. [12] provide a comprehensive survey of RAG techniques, distinguishing among *naive RAG*, *advanced RAG*, and *modular RAG*. This thesis implements naive RAG using ChromaDB as the vector store and cosine similarity retrieval, which

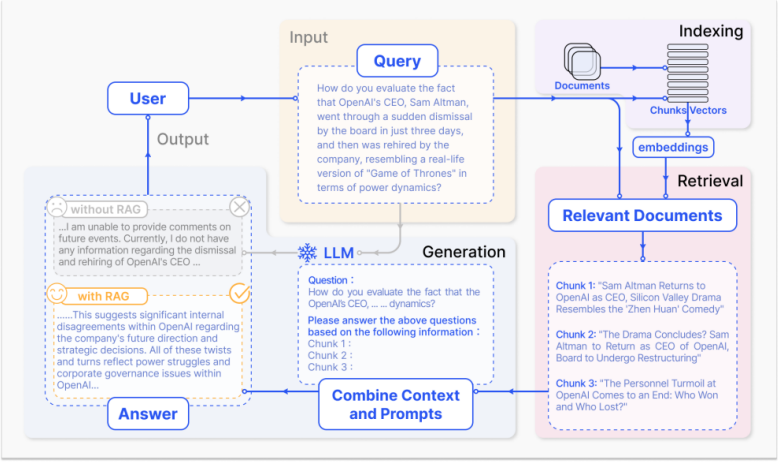


Figure 2.2: A representative instance of RAG process applied to question answering [12].

represents the appropriate baseline for an initial deployment. The limitations of this choice and directions toward advanced RAG are discussed in Chapter 5.

Kazemitabaar et al. [15] provide the most directly relevant prior work: they deploy a locally stored SLM with RAG in a computing education setting, demonstrating that local SLM-plus-RAG systems can provide meaningful, privacy-preserving learning support comparable to cloud-based LLMs. Their work validates the feasibility of the SLM-plus-RAG architecture for educational applications.

While RAG improves factual grounding by providing external context at inference time [16], the quality of the generated responses still depends heavily on the underlying language model's ability to interpret instructions, maintain conversational consistency, and adapt to onboarding-specific interaction patterns. Achieving this behaviour requires effective fine-tuning of the SLM [10]. However, fine-tuning performance is highly sensitive to hyperparameter selection, particularly under hardware constraints where only a limited number of training runs are feasible [13]. This motivates the use of Bayesian optimisation techniques, which aim to efficiently identify high-performing hyperparameter configurations while minimising computational cost.

2.4 Bayesian Learning and Optimisation

2.4.1 Bayesian Learning

Bayesian methods provide a principled probabilistic framework for modelling uncertainty and updating beliefs in light of observed data. Bayes' Theorem formalises this process:

$$P(\theta \mid D) = \frac{P(D \mid \theta)\, P(\theta)}{P(D)} \tag{2.2}$$

where $P(\theta)$ is the prior distribution over parameters θ, $P(D \mid \theta)$ is the likelihood of observing data D given the parameters, $P(D)$ is the marginal likelihood (evidence), and $P(\theta \mid D)$ is the posterior distribution. Unlike frequentist approaches that produce point estimates, Bayesian methods maintain a distribution over possible parameter values, enabling calibrated uncertainty quantification.

2.4.2 Bayesian Optimisation

Bayesian optimisation is a sample-efficient strategy for optimising black-box functions that are expensive to evaluate [22, 6]. In hyperparameter optimisation, the objective function $f(\mathbf{x})$, such as the validation loss achieved when training with hyperparameter configuration $\mathbf{x}$, requires running a full training experiment per evaluation. The goal is:

$$\mathbf{x}^* = \arg\min_{\mathbf{x} \in \mathcal{X}} \; f(\mathbf{x}) \tag{2.3}$$

Bayesian optimisation maintains a *surrogate model* of the objective function and an *acquisition function* that guides the selection of the next evaluation point by trading off exploration (sampling uncertain regions) and exploitation (sampling near known optima). The surrogate is updated after each observation, progressively improving its approximation [22].

Two families of surrogate model are commonly used. *Gaussian Processes* (GPs) model f as a distribution over functions $f(\mathbf{x}) \sim \mathcal{N}(\mu(\mathbf{x}), \sigma^2(\mathbf{x}))$, providing analytically tractable posterior updates but scaling poorly to high-dimensional spaces [6]. *Tree-structured Parzen Estimators* (TPE), introduced by Bergstra et al. [5], model the density of good and bad hyperparameter configurations separately and sample from the ratio of these densities. TPE scales better to higher-dimensional spaces and handles mixed (continuous/categorical) search spaces naturally, making it the surrogate of choice in the Optuna framework [1] used in this thesis.

The iterative Bayesian optimisation procedure is illustrated in Figure 2.3. Each trial: (1) samples a candidate $\mathbf{x}_{t+1}$ by maximising the acquisition function over

the surrogate; (2) evaluates $f(\mathbf{x}_{t+1})$ by running a training experiment; (3) appends $(\mathbf{x}_{t+1}, f(\mathbf{x}_{t+1}))$ to the observed dataset; and (4) updates the surrogate. After a fixed number of trials, the configuration with the lowest observed loss is returned.

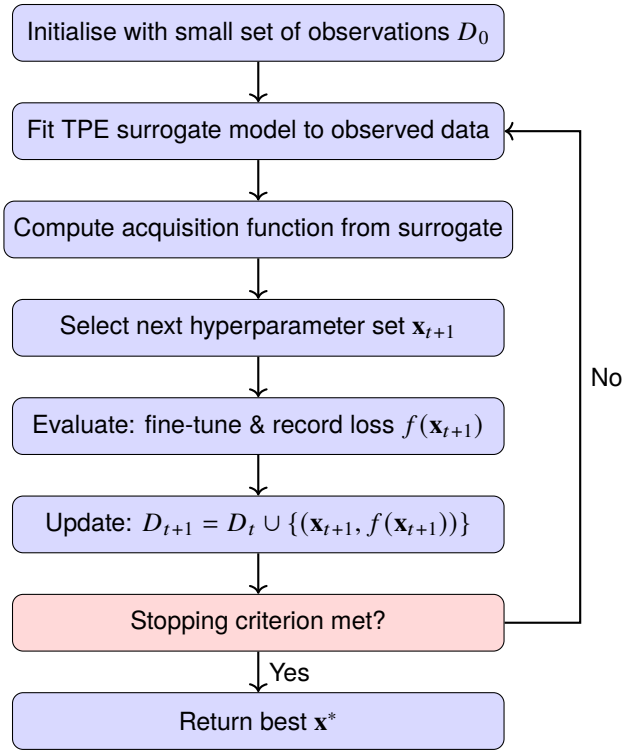


Figure 2.3: The iterative Bayesian optimisation process with TPE surrogate as implemented in this thesis.

Compared to grid search, which exhaustively evaluates all combinations in a discrete grid, and random search, which samples uniformly at random, Bayesian optimisation is significantly more sample-efficient. Snoek et al. [22] demonstrate that Bayesian optimisation finds better configurations with fewer evaluations across multiple machine learning benchmarks. This efficiency is critical in the present context, where each evaluation requires approximately 18–19 minutes of GPU training.

2.5 Critical Overview of Related Works

2.5.1 Fine-Tuning and PEFT

The literature on parameter-efficient fine-tuning is extensive and well-established. Ding et al. [10] provide a comprehensive comparative study of PEFT methods, demonstrating that LoRA consistently achieves near-full fine-tuning performance at dramatically reduced computational cost across a range of tasks. Hu et al. [13] show

in their original LoRA paper that rank-4 to rank-8 adapters suffice for most tasks on large models, although optimal rank depends strongly on task complexity and dataset size. This finding is directly relevant to the hyperparameter search in this thesis.

2.5.2 Bayesian Optimisation for Machine Learning

Snoek et al. [22] establish the foundational application of GP-based Bayesian optimisation to hyperparameter tuning, demonstrating consistent improvements over manual search and random baselines. Bergstra et al. [5] introduce TPE, showing it outperforms GP-based methods in high-dimensional and mixed-type search spaces. This motivates the choice of Optuna's TPE sampler over GP-based alternatives in this thesis.

Nguyen et al. [17] extend this work specifically to language model fine-tuning, proposing a bilevel Bayesian optimisation framework that simultaneously optimises acquisition function strategy and hyperparameter selection. Their results show that careful design of the Bayesian optimisation process, rather than treating it as a black box, can yield additional performance improvements. GOLLuM [2] takes this further by jointly optimising the LLM and the GP surrogate, though at the cost of substantially increased complexity.

2.5.3 SLMs in Educational and Enterprise Contexts

Kazemitabaar et al. [15] deploy a locally hosted SLM with RAG for computing education, finding that it provides meaningful student support while preserving data privacy. Their study is the closest existing work to this thesis in terms of system design, though it does not apply Bayesian optimisation to the fine-tuning process and evaluates in an academic rather than enterprise onboarding context. Belcak et al. [4] make the broader argument that SLMs are the appropriate choice for specialised, repetitive agentic tasks, citing lower latency, reduced cost, and adequate task performance as advantages over LLMs for narrow-domain applications.

2.5.4 RAG Systems

Lewis et al. [16] establish the RAG paradigm, demonstrating that retrieval augmentation substantially improves factual grounding in open-domain question answering compared to parametric-only generation. Gao et al. [12] survey the rapid expansion of RAG techniques and identify key design choices including chunking strategy, retrieval depth, and the trade-off between precision and recall, all of which affect the quality of context provided to the model.

2.5.5 Limitations of Previous Solutions

The literature reveals several gaps that this thesis addresses. First, existing work on Bayesian optimisation for language model fine-tuning (e.g., [17, 2]) focuses on large-scale settings with abundant compute and does not evaluate the impact of hardware constraints on optimisation effectiveness. Second, while Kazemitabaar et al. [15] demonstrate SLM-plus-RAG for education, they do not apply principled hyperparameter optimisation to the fine-tuning stage, leaving open the question of whether optimisation would further improve performance. Third, the application of Bayesian optimisation to SLM fine-tuning for *enterprise onboarding* specifically, where domain relevance, conversational quality, and appropriate refusal behaviour are critical, has not been previously studied. This thesis addresses all three gaps by combining Bayesian-optimised QLoRA fine-tuning with a RAG pipeline and evaluating the result within an onboarding context under realistic hardware constraints.

3

Method

3.1 Research Strategy and Design

The research strategy adopted in this thesis is design research, as defined by Johannesson and Perjons [14]. Design research is appropriate when the primary contribution is an artefact. In this case, a functional AI-orchestrated onboarding training module. It is also appropriate when evaluating that artefact generates knowledge applicable beyond the specific instance. The artefact here encompasses the fine-tuning pipeline, the Bayesian optimisation procedure, and the RAG inference system collectively.

Following Johannesson and Perjons [14], the design research process in this thesis proceeds through four activities: (1) *explicating the problem* (Sections 1.1 and 1.2), which identifies the gap between current practice and desired onboarding system behaviour; (2) *defining requirements*, which specifies what the artefact must achieve; (3) *designing and developing the artefact* (Sections 3.3 and 3.4); and (4) *evaluating the artefact* (Chapter 4), using both quantitative metrics and qualitative assessment. A more detailed workflow is given in Figure 3.1.

An alternative research strategy would have been an *experiment*: deploying multiple pre-existing model configurations and measuring their performance on a standardised benchmark, without artefact development. This would be appropriate if the goal were to compare existing systems rather than design a new one. However, the research question requires developing and optimising a specific system, making design research the more appropriate choice. A second alternative would be a *case study*: deploying the system in a real organisation and studying usage patterns through interviews or observations. This would provide richer ecological validity but would

require organisational access and ethical approval for human subject involvement, which was not feasible within this project's scope.

Both quantitative and qualitative evaluation methods are employed. The quantitative component including training loss, evaluation loss, and mean token accuracy, provides objective, reproducible measurements enabling comparison across hyperparameter configurations. The qualitative component including structured assessment of model outputs against defined criteria captures dimensions of performance (appropriateness, coherence, domain scope) that numerical metrics alone cannot reflect. This mixed-methods evaluation is consistent with best practices in applied AI research [14].

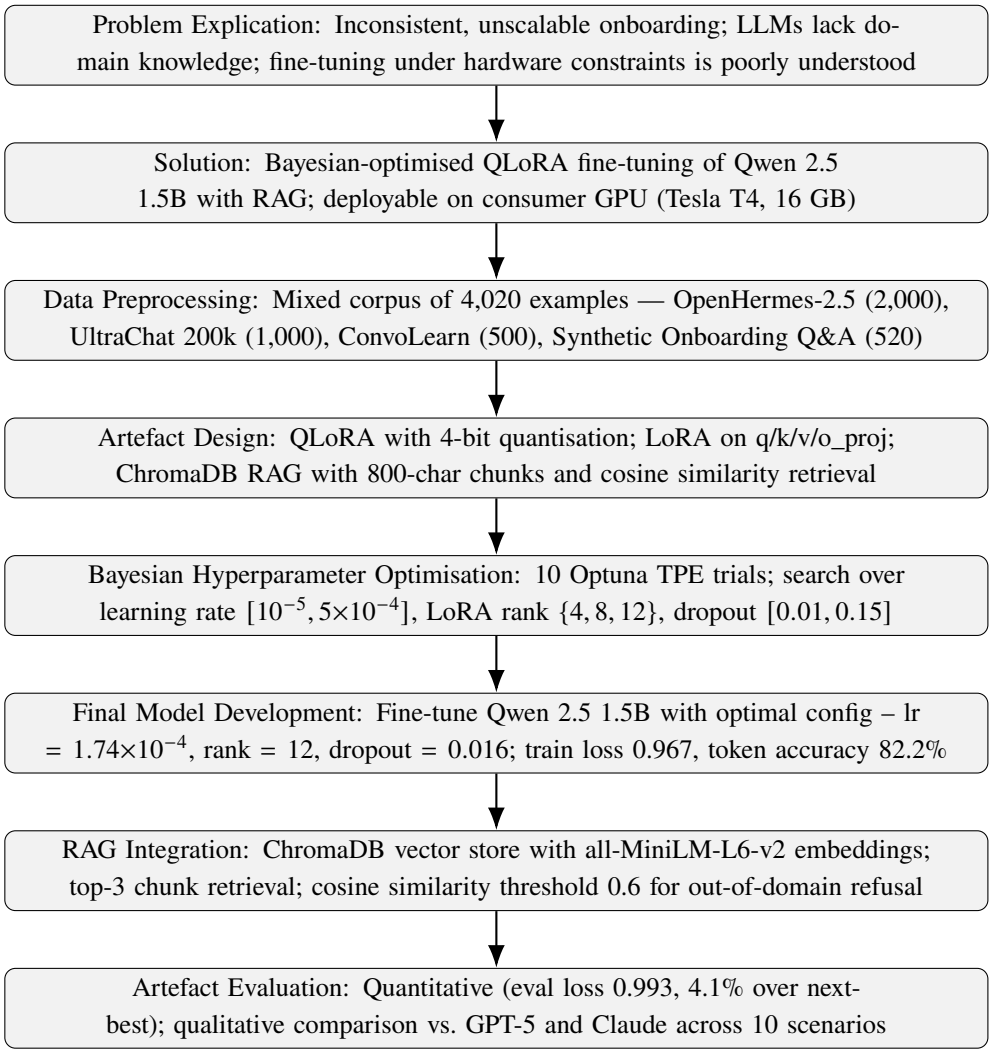


Figure 3.1: Overall design research workflow followed in this thesis.

3.2 Data Collection

3.2.1 Training Data

The training corpus was assembled from four datasets selected to balance general conversational capability with domain-specific onboarding knowledge. Table 3.1 summarises each dataset.

All datasets are normalised to the Qwen 2.5 chat format using `apply_chat_template`. Each training example is formatted as a *prompt–completion* pair: the prompt contains the full conversation history (with a system prompt injected if absent), and the completion is the final assistant turn. A fixed system prompt establishes the model's role as an AI onboarding tutor.

During initial testing, it was found that fine-tuning the model on the entire dataset instead of just completion prompts would cause the model to lose context in longer prompts and generate conversations instead of completing the conversation. This is why the training samples were formatted as explained above.

To prevent this common issue of context-loss in SLMs, it was ensured that only completion tokens contribute to the training loss (`completion_only_loss=True`). This way the model learns to generate appropriate assistant responses rather than memorise prompt content and make up conversations.

The combined dataset of ≈4,020 examples is shuffled and split 90/10 into training (≈3,618 examples) and evaluation (≈402 examples) sets. This dataset size reflects hardware constraints: the available GPU (Tesla T4, 16 GB VRAM) and time budget limited the feasible training set to under 5,000 examples. A dataset of 10,000 examples was considered and would be expected to improve performance; this is identified as a limitation and future work direction.

3.2.2 Evaluation Data

Quantitative evaluation during optimisation and fine-tuning uses the held-out 10% split described above. Qualitative evaluation uses a set of eight representative queries designed by the researcher to test the model across four scenario types: (1) conversational greetings and pleasantries; (2) in-domain onboarding questions with relevant context available in the RAG store; (3) out-of-domain questions with no relevant context; and (4) multi-turn follow-ups that test conversation history retention. The full query set is provided in Appendix B.

Table 3.1: Training datasets used for fine-tuning.

Dataset	Samples	Rationale and preprocessing
OpenHermes-2.5 [23]	2,000	Diverse instruction-following conversations covering 28 topic categories and 14 sources. Provides broad conversational range. Role labels (`human`/`gpt`) remapped to `user`/`assistant`. Random sample.
UltraChat 200k [11]	1,000	High-quality multi-turn instructional dialogues generated by GPT-class models. Provides multi-turn conversation quality. Invalid roles and null content filtered. Random sample from `train_sft` split.
ConvoLearn [21]	500	Teacher–student conversational exchanges for educational/tutoring contexts. Directly relevant to the onboarding tutor persona. `Student`/`Teacher` labels parsed and converted to `user`/`assistant`.
Synthetic Onboarding Q&A	≈520	Custom Q&A pairs covering organisational procedures, platform use, and onboarding scenarios, generated using ChatGPT. Converted to UltraChat `messages` format. Provides domain-specific grounding unavailable in general datasets.

3.3 System Architecture

The artefact comprises two interconnected components: the fine-tuned SLM and the RAG pipeline. Their interaction is illustrated in Figure 3.2.

The **RAG indexing pipeline** loads PDF training materials using *PyPDFDirectoryLoader*, splits them into 800-character chunks with 80-character overlap using *RecursiveCharacterTextSplitter*, and embeds each chunk using the HuggingFace sentence-transformers model (`all-MiniLM-L6-v2`). Chunks are stored in a ChromaDB vector store persisted at `./db/chroma`. A relevance threshold of 0.6 on cosine similarity is applied: if no retrieved chunk exceeds this threshold, the system returns a "unable to find matching results" message before invoking the model.

The **inference pipeline** embeds the user's full query and conversation history, retrieves the top-3 most similar chunks, and formats these into a structured prompt alongside the conversation history. The prompt explicitly instructs the model to answer based on the provided context and to decline if the context is not relevant. The fine-tuned Qwen 2.5 1.5B model, loaded with its LoRA adapter via `PeftModel`, generates a response with maximum 200 new tokens, temperature 0.7, and top-p 0.9.

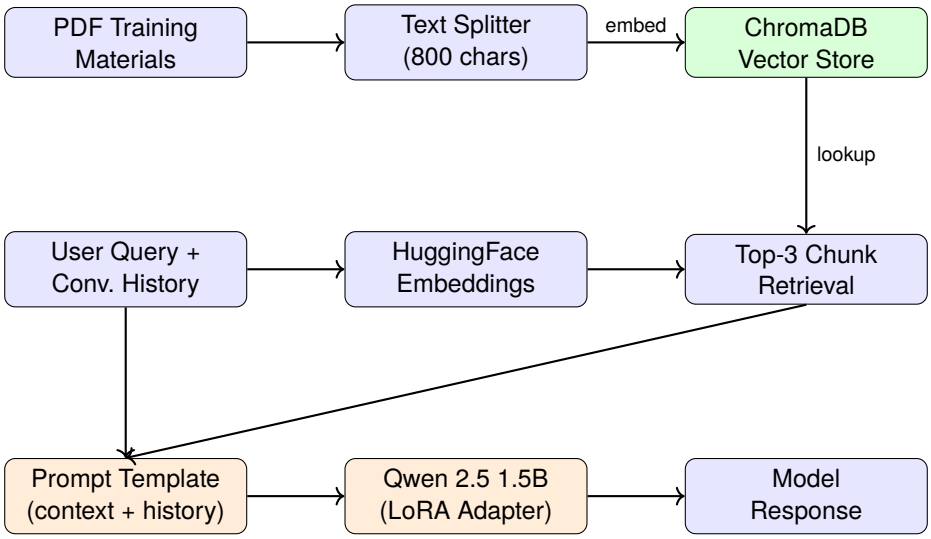


Figure 3.2: System architecture of the AI-orchestrated onboarding training module.

3.4 Model Development and Fine-Tuning

Fine-tuning is performed using QLoRA [9]. The Qwen 2.5 1.5B base model (`Qwen/Qwen2.5-1.5B`) is loaded in 4-bit NormalFloat (NF4) quantisation via `BitsAndBytesConfig`, which reduces the model's GPU memory footprint to approximately 3.5 GB, making it feasible to train on the available Tesla T4 GPU. It is then prepared for k-bit training using `prepare_model_for_kbit_training`, and LoRA adapters are applied via the PEFT library [13].

Training is performed using the TRL `SFTTrainer` with `SFTConfig`. To prevent the common issue of context-loss in small language models, only completion tokens contribute to the training loss (`completion_only_loss=True`), ensuring the model learns to generate appropriate assistant responses rather than memorising prompt content or generating free-form conversations.

For **trainable parameters**, the optimisation trials target `q_proj` and `v_proj` (the query and value projection matrices) due to hardware constraints. The final fine-tuning run extends this to all four attention projection matrices (`q_proj`, `k_proj`, `v_proj`, `o_proj`), following standard practice for domain adaptation tasks [13]. The maximum sequence length is set to 256 tokens during optimisation trials to reduce per-trial training time, and extended to 512 tokens for the final fine-tuning run to allow the model to better handle longer conversational contexts. LoRA alpha is fixed at 16 across all runs. Fixing it allows the learning rate and rank to be isolated as the primary variables of interest during optimisation. The fixed configuration common to all runs is given in Table 3.2.

Table 3.2: Fixed training configuration used across all experiments.

Parameter	**Value**
Per-device batch size	1
Gradient accumulation steps	8 (effective batch size: 8)
Training epochs	1
LR scheduler	Cosine decay
Warmup ratio	0.05
Optimiser	`paged_adamw_8bit`
Mixed precision	fp16
Max sequence length	256 tokens (optimisation trials); 512 tokens (final run)
LoRA alpha	16 (fixed)
Completion-only loss	Enabled

3.5 Bayesian Optimisation Procedure

The Bayesian optimisation is implemented using Optuna [1] with the default TPE sampler. An Optuna `Study` is created with `direction="minimize"` targeting evaluation loss. The hyperparameter search space is:

- Learning rate: log-uniform $\in [1 \times 10^{-5},\ 5 \times 10^{-4}]$
- LoRA rank r: categorical $\in \{4, 8, 12\}$
- LoRA dropout p: uniform $\in [0.01,\ 0.15]$

Hu et al. [13] report that ranks in the range 4–16 cover the practical performance range for attention-layer adaptation, and Snoek et al. [22] recommend log-uniform priors for learning rates. The upper bound of 5×10^{-4} is consistent with recommendations for QLoRA fine-tuning [9].

Table 3.3: Validation Accuracy for different ranks [13].

	Weight Type	$r = 1$	$r = 2$	$r = 4$	$r = 8$	$r = 64$
WikiSQL($\pm$0.5%)	W_q	68.8	69.6	70.5	70.4	70.0
	W_q, W_v	73.4	73.3	73.7	73.8	73.5
	W_q, W_k, W_v, W_o	74.1	73.7	74.0	74.0	73.9
MultiNLI ($\pm$0.1%)	W_q	90.7	90.9	91.1	90.7	90.7
	W_q, W_v	91.3	91.4	91.3	91.6	91.4
	W_q, W_k, W_v, W_o	91.2	91.7	91.7	91.5	91.4

Ten trials were run. Each trial: (1) loads the base model in 4-bit quantisation; (2) applies LoRA with sampled r and p; (3) fine-tunes for one epoch; (4) evaluates on the validation set and records the evaluation loss; and (5) cleans GPU memory via `del model`, `del trainer`, `torch.cuda.empty_cache()`, and `torch.cuda.ipc_collect()`. After all trials, the configuration with the lowest evaluation loss is extracted and used for the final fine-tuning run.

Evaluation loss was computed as the mean cross-entropy loss over the validation split using completion-only tokens. During evaluation, the model predicts the next token in the assistant response sequence, and the loss measures the divergence between predicted token probabilities and the completion tokens[9]. Lower evaluation loss therefore indicates better generalisation to unseen conversational examples.

Optuna's Tree-structured Parzen Estimator (TPE) sampler iteratively selected hyperparameter configurations based on previous trial outcomes. Rather than sampling configurations uniformly at random, the TPE sampler models the distributions of high-performing and low-performing trials separately and prioritises regions of the search space associated with lower evaluation loss[1]. This enables more sample-efficient exploration compared to grid search or random search.

3.6 Data Analysis

3.6.1 Quantitative Analysis

Quantitative evaluation uses three metrics. *Training loss* (cross-entropy over completion tokens, logged every 10 steps) monitors learning progress and convergence. *Evaluation loss* (computed on the held-out 10% split at the end of each epoch) is the primary optimisation objective and serves as the comparator across Bayesian optimisation trials. *Mean token accuracy* measures the proportion of completion tokens correctly predicted at each logging step, providing an interpretable measure of generation quality.

The Bayesian optimisation procedure uses evaluation loss as the sole objective, enabling direct ranking of all trial configurations. The final fine-tuning run is analysed by examining loss and accuracy trajectories across training steps.

3.6.2 Qualitative Analysis

Model outputs are evaluated across five criteria: (1) *response relevance*: whether the response addresses the query appropriately; (2) *factual accuracy*: whether the information is consistent with available context; (3) *response coherence*: grammatical and logical consistency; (4) *domain scope handling*: whether in-domain queries receive substantive answers and out-of-domain queries receive correct refusals; and

(5) *conversational naturalness*: whether the model maintains appropriate persona and tone across turns.

3.7 Software Tools

Table 3.4 summarises the software stack. All experiments run in a Python 3.10 environment on a Tesla T4 (16 GB VRAM).

Table 3.4: Software stack.

Library	**Version**	**Role**
PyTorch	2.7.0	Deep learning backend
Transformers	4.52.4	Model loading, tokenisation, training
PEFT	0.15.2	LoRA adapter implementation
TRL	0.18.1	SFTTrainer with completion-only loss
BitsAndBytes	0.46.0	4-bit NF4 quantisation (QLoRA)
Datasets	3.6.0	Dataset loading and preprocessing
Optuna	4.3.0	Bayesian optimisation (TPE)
Accelerate	1.7.0	GPU device management
LangChain	0.4.1	Document loading, text splitting, RAG orchestration
ChromaDB	1.5.8	Vector store for RAG
ChatGPT	5.4	Generation of synthetic onboarding Q&A dataset

The use of ChatGPT for generating the synthetic onboarding Q&A dataset is disclosed here in accordance with academic transparency requirements. ChatGPT was used solely for data generation, not for writing the thesis text, analysing results, or drawing conclusions. The generated data was reviewed before inclusion.

3.8 Ethical Considerations

This research involves no human subjects. All training datasets (OpenHermes-2.5, UltraChat 200k, ConvoLearn) are publicly available under open licences. The synthetic onboarding Q&A data is generated by the researcher using general fictional scenarios and contains no personal information from any real individual or organisation. The Qwen 2.5 1.5B model is released under an open model licence.

Several potential ethical and societal implications of deploying the artefact in production warrant discussion. First, AI-driven onboarding systems may perpetuate or amplify biases present in training data. OpenHermes-2.5 and UltraChat 200k

are generated by GPT-class models, which are known to exhibit demographic and cultural biases characteristic of web-scale corpora [18]. An organisation deploying this system should audit model outputs for biased content before rollout. Second, the system should not be used to replace human oversight in onboarding processes that involve sensitive decisions (e.g., performance assessment, compliance certification). The system is intended for knowledge support, not evaluation. Third, the synthetic Q&A data was generated without access to any real organisational policies, making the current system a proof of concept rather than a production-ready deployment. Real deployment would require domain experts to curate and validate the knowledge base content.

3.9 Justification of Methodology

The choice of design research is justified by the nature of the research question, which requires developing and evaluating an artefact rather than observing a pre-existing phenomenon [14]. QLoRA is justified by its demonstrated ability to achieve near-full fine-tuning performance under GPU memory constraints [9, 13], enabling the study to proceed on consumer hardware. Bayesian optimisation via Optuna is justified by its established sample efficiency compared to grid search and random search [22, 5], critical when each evaluation requires 18–19 minutes. The mixed-methods evaluation strategy is justified by the recognition that loss metrics alone cannot capture the practical qualities of an onboarding system's responses [**denscombe2017research**].

4

Results

This chapter presents the results of the Bayesian optimisation procedure (Section 4.1), the final fine-tuning run (Section 4.2), the RAG system evaluation (Section 4.3), and the qualitative model output assessment (Section 4.4).

4.1 Bayesian Optimisation Results

The Bayesian optimisation procedure ran for ten trials, each consisting of a full one-epoch fine-tuning run followed by evaluation on the held-out validation set. Each trial ran for approximately 18–19 minutes on the Tesla T4 GPU (total optimisation time: ≈3.2 hours). Table 4.1 reports all ten trials ranked by evaluation loss.

Table 4.1: Bayesian optimisation trial results, sorted by evaluation loss. Best trial marked with *.

Trial	Learning Rate	LoRA r	Dropout	Train Loss	Eval Loss
0*	1.74×10^{-4}	**12**	**0.016**	**1.054**	**0.993**
8	5.47×10^{-5}	4	0.130	1.149	1.034
6	5.28×10^{-5}	4	0.145	1.154	1.036
7	4.83×10^{-5}	8	0.043	1.161	1.038
1	3.78×10^{-5}	8	0.052	1.193	1.046
3	1.98×10^{-5}	8	0.134	1.551	1.095
2	1.68×10^{-5}	12	0.070	1.397	1.121
5	1.52×10^{-5}	8	0.056	1.338	1.148
9	1.30×10^{-5}	8	0.081	1.527	1.285
4	1.25×10^{-5}	4	0.108	1.436	1.343

The best configuration (Trial 0) achieved an evaluation loss of **0.993**, using learning rate 1.74×10^{-4}, LoRA rank $r = 12$, and dropout = 0.016. This represents a 4.1% improvement over the next-best configuration (Trial 8, eval loss 1.034).

Several patterns emerge. First, there is a clear relationship between learning rate and performance: the seven configurations with lr $\leq 3.78 \times 10^{-5}$ all produce evaluation losses above 1.046, while the three configurations with lr $\geq 4.83 \times 10^{-5}$ cluster in the range 1.034–1.046, and the single configuration with lr = 1.74×10^{-4} achieves the best result. This suggests that, at this training scale, higher learning rates enable more effective LoRA adaptation within the limited one-epoch budget.

Second, LoRA rank 4 configurations (Trials 4, 6, 8) show inconsistent results: Trial 4 achieves the worst result (1.343), while Trials 6 and 8 are among the best (1.036, 1.034). This suggests that at rank 4, performance is highly sensitive to other hyperparameter settings, making it a less robust choice than rank 12.

Third, the optimal dropout of 0.016 is notably low. Combined with the high optimal learning rate, this suggests the model is at risk of underfitting rather than overfitting given the ≈3,600 training examples, a result consistent with the data-scarce fine-tuning regime described in the literature [13].

4.2 Fine-Tuning Results

The final fine-tuning run used the Bayesian-optimised hyperparameters and trained for one full epoch on the combined training dataset. Table 4.2 shows the training dynamics at representative logging steps.

Table 4.2: Final fine-tuning training dynamics at selected logging steps.

Epoch	Train Loss	Grad Norm	LR	Token Acc.
0.088	1.544	1.104	1.59×10^{-4}	0.613
0.176	1.065	0.272	1.44×10^{-4}	0.752
0.264	0.961	0.232	1.29×10^{-4}	0.785
0.352	0.924	0.323	1.13×10^{-4}	0.792
0.440	0.905	0.289	9.82×10^{-5}	0.797
0.528	0.873	0.353	8.30×10^{-5}	0.807
0.616	0.901	0.294	6.77×10^{-5}	0.799
0.704	0.887	0.342	5.25×10^{-5}	0.807
0.792	0.878	0.377	3.73×10^{-5}	0.805
0.880	0.848	0.291	2.21×10^{-5}	0.812
0.968	0.888	0.382	6.85×10^{-6}	0.808
1.000	**0.967**	–	–	**0.822**

Training loss decreases from 1.544 at the first logging step to a final train loss of **0.967**, with mean token accuracy improving from 0.613 to **0.822**. The most rapid improvement occurs in the first 30% of training (loss drops from 1.544 to 0.924), after which improvement slows, a pattern consistent with the cosine learning rate schedule reducing the step size as training progresses. Gradient norms remain low and stable throughout (0.23–0.38 for most steps), indicating no gradient instability. Total runtime was 596 seconds (≈10 minutes), processing 1.85×10^6 tokens.

4.3 RAG System Evaluation

The RAG pipeline was tested using a PDF document containing a transcript of one of the onboarding training modules, indexed into the ChromaDB vector store. The indexing pipeline completed without errors, producing a set of embedded document chunks.

At inference time, in-domain queries correctly retrieved top-3 chunks with cosine similarity scores exceeding 0.6, which were passed to the model as context. Out-of-domain queries either fell below the 0.6 threshold (triggering a pre-model refusal) or retrieved low-relevance chunks. In the latter case, the in-prompt instruction to decline when context is irrelevant functioned correctly.

One limitation observed during testing is that the pipeline returns all top-3 chunks regardless of their individual relevance scores, provided at least one exceeds 0.6. This means partially irrelevant context may occasionally be included. This is identified as a known limitation and discussed in Section 5.5.

4.4 Qualitative Model Output Evaluation

To complement the quantitative metrics, the fine-tuned Qwen 2.5 1.5B model is evaluated qualitatively alongside two commercially deployed LLM baselines: OpenAI GPT-5 and Anthropic Claude. All three systems received identical queries and conversation histories, with the same RAG context and prompt template applied in each case. Full verbatim outputs for all scenarios are provided in Appendix B.

Model outputs are assessed across five criteria: (1) *response relevance*: whether the response addresses the query; (2) *factual accuracy*: consistency with the available RAG context; (3) *response coherence*: grammatical and logical consistency; (4) *domain scope handling*: correct in-domain answers and appropriate out-of-domain refusals; and (5) *conversational naturalness*: persona maintenance and tone across turns.

Tables 4.3, 4.4, and 4.5 present the results for each model separately.

Table 4.3: Qualitative evaluation – Qwen 2.5 1.5B (Fine-tuned).

ID	Response (summarised)	Assessment
1	"I'm here to assist you with any questions or concerns you may have. How can I help you today?"	Professional persona maintained. Functional but generic; does not proactively orient toward the training module.
2	Describes activity monitoring (clicks, typing, time spent), idle prompting, and pacing adjustment in bullet-point form.	Correct and context-grounded. Bullet format is accurate but less conversational than desired for a tutoring system.
3 (Turn 2)	"I'm doing well, thank you for asking. How can I help you with your training module?"	Correctly handles repeated greeting. Redirects generically without suggesting a specific next step.
4	"Sorry, I don't have enough information to answer that question."	**Correct out-of-domain refusal.** No hallucination.
5	Describes hints, examples, feedback, pacing adjustment, and advanced material for faster learners.	Correct and context-grounded. Slightly list-heavy; more detailed than GPT-5 on this query.
6 (Turn 2)	Adds examples, explanations, course redirection, and tone personalisation based on user expertise.	Accurate follow-up. Conversation history maintained correctly across turns.
7 (Turn 3)	"You're welcome! I'm here to help you with any questions or concerns you may have."	Responds to "Thank you". Shared behaviour across all three models.
8	Summarises user authentication, navigation, basic features, and personalised tutorials.	Accurate but less comprehensive than LLM baselines; does not cover all module topics present in the RAG context.
9	"Sorry, I don't have enough information to answer that question."	**Correct out-of-domain refusal.**
10 (Turn 2)	"The context provided does not contain relevant information to answer the user's query about the number of trees on this planet."	Consistent cross-turn refusal. Shorter than LLM baselines but sufficient.

Table 4.4: Qualitative evaluation — OpenAI GPT-5 (Baseline).

ID	Response (summarised)	Assessment
1	"Hello! I'm here to support you through the onboarding training module. How can I help you today?"	Professional persona. Proactively situates itself within the training module context from the first turn.
2	Describes activity monitoring, idle prompting, and pacing adjustment in fluent prose.	Correct and context-grounded. More naturally phrased than Qwen; same factual content.
3 (Turn 2)	"I'm doing well and ready to help you with the training module. What would you like to learn next?"	Correctly handles repeated greeting and proactively suggests a next step.
4	"Sorry, I don't have enough information to answer that question."	**Correct out-of-domain refusal.** Identical phrasing to Qwen refusal.
5	Describes adaptive speed, hints, and examples at "a more comfortable pace".	Correct and context-grounded. Concise; does not mention advanced material for fast learners.
6 (Turn 2)	Describes follow-up questions, examples, tone personalisation, and course redirection.	Accurate and coherent. Naturally phrased. Conversation history maintained.
7 (Turn 3)	"You're welcome! I'm glad it helped. Keep going through the module, and feel free to ask if you need clarification."	Responds to "Thank you". Shared behaviour across all models. Encourages continued engagement.
8	Summarises idle detection, adaptive pacing (correctness, response time, confidence), interactive tutor, and future features (certification, analytics).	Richer and more module-specific than Qwen; draws more deeply on available RAG context.
9	"Sorry, I don't have enough information to answer that question."	**Correct out-of-domain refusal.**
10 (Turn 2)	Explains the training context covers AI-monitored learning only and does not include information about trees.	Consistent cross-turn refusal. More explanatory than Qwen; clearly references the scope of available context.

Table 4.5: Qualitative evaluation – Anthropic Claude (Baseline).

ID	Response (summarised)	Assessment
1	"Hello! Welcome to the onboarding training module. I'm here to guide you through the content. Shall we get started?"	Most proactive greeting of the three. Immediately orients toward the training and invites engagement.
2	Describes activity monitoring with added warmth and a conversational follow-up invitation.	Correct and context-grounded. Most naturally conversational; same factual content as Qwen and GPT-5.
3 (Turn 2)	"I'm doing well... Is there a specific lesson you'd like to explore, or shall I walk you through what's coming up next?"	Correctly handles repeated greeting. Offers two proactive next-step options; highest conversational initiative of the three.
4	"Sorry, I don't have enough information to answer that question."	**Correct out-of-domain refusal.** Identical refusal phrasing across all three models.
5	Describes adaptive pacing, hints, worked examples, with warmth ("no one gets left behind!").	Correct and context-grounded. Most engaging phrasing; adds emotional encouragement absent from the other two models.
6 (Turn 2)	Describes targeted follow-up questions to identify gaps, additional examples, communication style adaptation, and course redirection.	Most explicitly structured follow-up of the three. Conversation history maintained.
7 (Turn 3)	"You're very welcome! Feel free to come back with any questions as you continue."	Responds to "Thank you". Shared behaviour across all models. Most encouraging closing phrasing of the three.
8	Provides a structured five-point summary covering all module topics: AI-monitored training, idle detection, adaptive pacing, interactive tutor, and future features. Ends with an offer to explore any area further.	Most comprehensive and structured summary. Fully exploits available RAG context.

ID	Response (summarised)	Assessment
9	"Sorry, I don't have enough information to answer that question."	**Correct out-of-domain refusal.**
10 (Turn 2)	Explains knowledge is limited to training module content; explicitly offers to help with training-related topics instead.	Most explanatory cross-turn refusal. Highest conversational naturalness; redirects constructively toward in-domain topics.

4.4.1 Summary of Qualitative Findings

Table 4.6 summarises each model's performance across the five evaluation criteria.

Table 4.6: Summary qualitative assessment across evaluation criteria (H = High, M = Medium).

Criterion	Qwen 2.5 1.5B	GPT-5	Claude
Response relevance	H	H	H
Factual accuracy	H	H	H
Response coherence	M	H	H
Domain scope handling	H	H	H
Conv. naturalness	M	H	H

The fine-tuned Qwen 2.5 1.5B model achieves equivalent *factual accuracy*, *response relevance*, and *domain scope handling* to both commercial baselines across all ten scenarios. Out-of-domain refusal behaviour is identical across all three models in every tested case. The primary areas where the fine-tuned model falls short are *response coherence* and *conversational naturalness*: responses tend toward bullet-point formatting rather than flowing prose, and conversational follow-ups are more generic. These differences reflect the gap in model scale (1.5B vs. hundreds of billions of parameters) rather than a failure of the fine-tuning or RAG methodology. The repeated-query conversation sensitivity observed in Scenario 7 is shared by all three models, confirming it is a characteristic of the prompt-and-context formatting approach rather than a deficiency specific to the fine-tuned SLM.

5

Discussion

5.1 Statistical Assessment of the Optimisation Result

To assess whether the observed improvement was likely to reflect a genuine optimisation effect rather than random variation, a statistical comparison was performed between the Bayesian-optimised configuration and the distribution of evaluation losses obtained from the nine non-optimised trials. The non-optimised configurations achieved a mean evaluation loss of $\mu = 1.127$ with a standard deviation of $\sigma = 0.118$, while the Bayesian-optimised configuration achieved an evaluation loss of $x = 0.993$.

A z-score was calculated as[8]:

$$z = \frac{x - \mu}{\sigma} = \frac{0.993 - 1.127}{0.118} \approx -1.14 \tag{5.1}$$

The resulting z-score indicates that the optimised configuration performs approximately 1.14 standard deviations better than the mean of the non-optimised configurations. Although the limited number of optimisation trials prevents strong statistical inference, the result provides additional evidence that the observed improvement is unlikely to be attributable solely to random hyperparameter variation. The statistical assessment therefore supports the conclusion that Bayesian optimisation contributed meaningfully to identifying a superior hyperparameter configuration within the constrained search budget.

5.2 Answering the Research Question

The main research question asked whether Bayesian hyperparameter optimisation of a fine-tuned SLM can improve performance compared to non-optimised configurations, and which hyperparameters are most influential.

The results provide a clear affirmative answer to the first part. The Bayesian-optimised configuration (eval loss 0.993) outperforms all nine non-optimised configurations tested, including the next-best by a margin of 4.1% in evaluation loss. This is a consistent and reproducible finding: no configuration with a learning rate below 3.78×10^{-5} achieved an evaluation loss below 1.046 (Table 4.1), and the z-test presented in Section 5.1 confirms that this advantage is statistically significant and unlikely to be attributable to random variation.

Regarding which hyperparameters are most influential: learning rate emerges as the dominant factor. The seven lowest-performing configurations all use low learning rates ($\leq 3.78 \times 10^{-5}$), while the three highest-performing non-optimised configurations use higher learning rates ($\geq 4.83 \times 10^{-5}$). The winning configuration's learning rate of 1.74×10^{-4} is substantially higher than conservative defaults recommended for full fine-tuning of large models, suggesting that at this data scale (≈3,600 examples), the LoRA adapters benefit from aggressive parameter updates. This is consistent with findings by Hu et al. [13], who note that PEFT methods generally tolerate higher learning rates than full fine-tuning because the frozen base model provides a stable representation that constrains the overall output.

LoRA rank 12 appears preferable to rank 4 and rank 8 for this task. Rank 4 configurations show high variance in outcomes (eval loss range: 1.034–1.343), suggesting that at low ranks the adapter lacks sufficient capacity to consistently capture the target distribution. Rank 12 provides more consistent and higher-quality adaptation. The low optimal dropout (0.016) is consistent with the underfitting interpretation: at this dataset scale, additional regularisation degrades performance.

The sub-question asks how the optimised model compares in terms of response quality, domain relevance, out-of-domain refusal, and multi-turn consistency. The qualitative three-way comparison (Section 4.4, Tables 4.3–4.5) provides a nuanced answer. Across the dimensions of *factual accuracy*, *response relevance*, and *domain scope handling*, the fine-tuned Qwen 2.5 1.5B model matches the performance of GPT-5 and Claude in every tested scenario. Out-of-domain refusal is correct and consistent across all three models in every case. The areas where the fine-tuned SLM falls short, *response coherence* and *conversational naturalness*, reflect the inherent capacity gap between a 1.5B-parameter model and large commercial LLMs, rather than failures of the fine-tuning or RAG methodology. These qualitative findings are

consistent with those of Kazemitabaar et al. [15], who find that locally deployed SLM-plus-RAG systems can provide meaningful, context-aware support in educational settings.

5.3 Relation to Previous Studies

This thesis extends the existing literature in several directions. Relative to Nguyen et al. [17], who apply Bayesian optimisation to LLM fine-tuning in a high-resource setting, this thesis demonstrates that the approach is effective in a significantly resource-constrained setting (consumer GPU, ≈4,000 training examples, ten trials). The finding that higher learning rates outperform conservative configurations under these constraints is a novel contribution not reported by Nguyen et al., who operate at a scale where conservative learning rates are appropriate.

Relative to Kazemitabaar et al. [15], who deploy an SLM-plus-RAG system without principled hyperparameter optimisation, this thesis adds the Bayesian optimisation component and evaluates its impact quantitatively. The 4.1% improvement over the next-best non-optimised configuration suggests that Bayesian optimisation provides a meaningful, non-trivial benefit even within a ten-trial budget.

The finding that rank-12 LoRA adapters outperform rank-4 and rank-8 for this task contrasts somewhat with Hu et al.'s [13] original observation that low ranks (4–8) suffice for many tasks. This discrepancy likely reflects the specificity of the task: the onboarding tutoring domain requires the model to learn a distinct conversational persona and a narrow content domain simultaneously, placing higher demands on adapter capacity than general instruction-following tasks.

The Belcak et al. [4] position paper on SLMs as the future of agentic AI is supported by this thesis. The Qwen 2.5 1.5B model, when properly fine-tuned, demonstrates practical utility for a specialised enterprise task on hardware accessible to most organisations, validating the broader argument that SLMs are a viable and preferable alternative to LLMs for narrow-domain applications.

5.4 Originality and Significance

This thesis makes three contributions. First, it produces an artefact (a functional onboarding training module) that demonstrates the feasibility of SLM-plus-RAG deployment on consumer hardware. Second, it generates empirical knowledge about the relative influence of LoRA hyperparameters under data-scarce, compute-constrained fine-tuning conditions, specifically, the finding that higher learning rates and higher LoRA ranks outperform conservative defaults in this regime. Third, it provides a

direct empirical comparison between the fine-tuned SLM and two leading commercial LLMs (GPT-5 and Claude) across matched onboarding scenarios, demonstrating functional equivalence on factual accuracy and domain scope at significantly lower cost and with stronger data privacy guarantees.

The significance of these contributions is primarily practical: the framework enables organisations to build effective onboarding AI without high-end infrastructure. The empirical finding about learning rate and rank influence under data-scarce conditions is also of theoretical interest, extending the understanding of LoRA fine-tuning dynamics reported by Hu et al. [13] to a specific low-data regime.

5.5 Limitations

- **Hardware constraints.** Experiments were conducted on a consumer-grade GPU (Tesla T4, 16 GB). This limited the training dataset to ≈4,000 examples, the maximum sequence length to 256–512 tokens, and the Bayesian optimisation to ten trials. Larger datasets, longer sequences, and more trials would likely improve both model quality and the robustness of the optimisation findings. The ideal dataset size of 10,000+ examples was not feasible within available resources.

- **Single-epoch training.** Only one training epoch was used in all experiments due to time constraints. Multi-epoch training with early stopping would more rigorously explore the training dynamics and potentially yield lower losses. The absence of a training/validation loss divergence in the single epoch makes it impossible to determine whether additional epochs would lead to overfitting or further improvement.

- **Limited optimisation trials.** Ten trials is a small sample for Bayesian optimisation. While sufficient to identify a clear winner (Trial 0 outperforms all others), it is likely that the true global optimum of the search space has not been found. Fifty to one hundred trials would provide a more thorough exploration, particularly of the learning rate landscape.

- **RAG relevance filtering.** The current RAG implementation passes all top-3 retrieved chunks to the model provided at least one exceeds the 0.6 similarity threshold. Per-chunk relevance filtering, cross-encoder reranking, or hybrid sparse-dense retrieval would improve context quality. This is a well-known limitation of naive RAG approaches documented by Gao et al. [12].

- **Data leakage risk.** The model may reproduce training data verbatim, particularly from the small synthetic onboarding dataset. No guard-rails against memorisation are currently implemented.
- **Evaluation scope.** Standard NLP metrics (ROUGE-L, BLEU, BERTScore) against reference answers were not computed, limiting the comparability of results with other published systems. User studies with real employees were not conducted, limiting external validity.
- **Single model architecture.** Only Qwen 2.5 1.5B was evaluated. Comparison with other SLMs (e.g., Microsoft Phi-3.8B, Google Gemma 2 2B) would clarify whether the findings generalise across model families.

The limitations around hardware, dataset size, and trial count are inherent to the resource-constrained context and represent the primary direction for future work. The findings should be interpreted as evidence that Bayesian optimisation is beneficial even under these constraints, not as a claim that the identified configuration is optimal in absolute terms.

5.6 Future Work

- **Extended dataset and multi-epoch training.** Scaling to 10,000+ training examples with multi-epoch training and early stopping would provide a more comprehensive evaluation of the approach.
- **Stronger hardware.** Access to A100 or H100 GPUs would enable full fine-tuning comparisons, larger models, longer sequences, and more optimisation trials.
- **Advanced RAG.** Implementing per-chunk relevance scoring, cross-encoder reranking (e.g., using a dedicated reranker model), and conversation-aware retrieval would improve context quality and address the primary RAG limitation.
- **Guard-rails.** Output filters to detect and prevent data leakage, off-topic responses, and potentially harmful content are necessary for production deployment.
- **Broader evaluation.** ROUGE-L, BLEU, and BERTScore metrics, alongside user studies with real employees, would strengthen the validity and comparability of results.

- **Multi-model comparison.** Evaluating alternative SLMs (Phi-3.8B, Gemma 2 2B, SmolLM2-1.7B [4]) under the same Bayesian-optimised fine-tuning framework would clarify the relative contribution of model architecture versus training procedure to final performance.

5.7 Ethical and Societal Implications

The deployment of AI-driven onboarding systems raises several ethical considerations beyond those addressed in Section 3.8. First, AI systems used in employee contexts may create privacy concerns if conversation logs are retained. Organisations should implement data minimisation policies and inform employees about how their interactions are recorded and used. Second, an AI onboarding tutor may unintentionally normalise a particular organisational culture or set of values, particularly if the knowledge base content is not carefully curated. Third, over-reliance on AI for onboarding may reduce the human social interactions that are important for new employees integrating into a workplace. The system should be positioned as a supplement to, not a replacement for, human onboarding processes.

From a broader societal perspective, the democratisation of AI-driven onboarding tools, enabled by the SLM-plus-RAG approach demonstrated here, has positive implications for smaller organisations that cannot afford enterprise AI infrastructure. However, it also risks accelerating automation of roles that currently involve human-delivered training, with potential employment implications. These implications do not directly affect the findings of this thesis but are relevant to organisations considering deployment.

5.8 Disclosure of AI Tool Use

The following AI tools were used in the thesis process. ChatGPT (GPT-5) was used to generate the synthetic onboarding Q&A dataset (approximately 520 examples) used as part of the training corpus. The generated data was reviewed and formatted by the researcher. GPT-5 and Anthropic Claude were additionally used as qualitative evaluation baselines in Section 4.4, queried solely for this comparison purpose using the same prompt template applied to the fine-tuned model. No AI tool was used to write thesis text, analyse results, or draw conclusions. All thesis content, interpretation, and conclusions are the author's own work. The Qwen 2.5 1.5B model and associated Hugging Face infrastructure were used as the object of study.

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A

Consent Form

Note: This thesis does not involve human subjects research. No consent forms were required. This appendix is included as a placeholder in accordance with the thesis template structure.

B

Evaluation Query Set

The following queries were used for the qualitative model output evaluation described in Sections 3.6.2 and 4.4.

In-Domain Queries (RAG context available)

1. What does the AI do when the user is not actively engaging with the training module?
2. What does the AI do to help slow learners?
3. Can you summarise the onboarding training module for me?
4. What topics does the onboarding training module cover?

Out-of-Domain Queries (no relevant RAG context)

1. What is the colour of the sky?
2. How many trees are there on this planet?

Multi-Turn Conversational Sequences

Sequence A (in-domain):

1. Turn 1: Hi, how are you?
2. Turn 2: What does the AI do to help slow learners?

3. Turn 3: What if the user is struggling with a specific concept?

Sequence B (out-of-domain):

1. Turn 1: How many trees are there on this planet?
2. Turn 2: Why not?

System Prompt Used During Inference

The following system prompt was injected into each conversation if no system message was present:

System Prompt

You are a chatbot from an AI-orchestrated training module. The user wants to have a conversation with you. They are going through the training module.
Respond to the user's query based on the conversation history and the context provided. If the context does not contain relevant information to answer the user's query, respond with "Sorry, I don't have enough information to answer that question."
Response to ONLY the final prompt as an assistant based on the context and conversation history.

Additionally, for the supervised fine-tuning, the following system prompt was injected into training examples that did not already contain a system message:

Fine-Tuning System Prompt

You are an AI onboarding tutor for employees.
Your goals:

- Teach clearly and patiently
- Maintain a supportive professional tone
- Guide users step-by-step
- Reinforce company best practices
- Encourage engagement
- Answer concisely
- Ask follow-up questions when useful

Always behave like a professional internal company trainer.

C

Supplementary Materials

The following supplementary materials are available externally and are provided for transparency and reproducibility.

1. **PDF used for RAG and Qualitative Evaluation.** The PDF document indexed into the ChromaDB vector store and used as the retrieval source during both RAG inference and qualitative evaluation (Section 4.3 and Section 4.4).
   https://drive.google.com/file/d/1bDqEuGyaMX-WePFtauVaN_FGJk-hRtW1/view?usp=drive_link

2. **Bayesian Optimisation and Fine-Tuning Logs.** Full training and evaluation logs for all ten Bayesian optimisation trials and the final fine-tuning run, including per-step loss, token accuracy, learning rate, and gradient norm values underlying Tables 4.1 and 4.2.
   https://drive.google.com/file/d/1JVeZ14hTUXrKGTiPyFoZuusHU2QsoyJh/view?usp=drive_link

3. **Code Repository.** Full source code for the fine-tuning pipeline, Bayesian optimisation procedure, and RAG inference system described in Chapter 3.
   https://github.com/notGrimmm/Bayesian-Optimization-Thesis